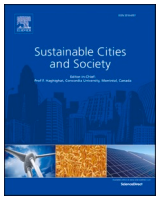
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Optimal sizing of standalone rural microgrid for sustainable electrification with renewable energy resources

Md. Mustafa Kamal [a,*], Imtiaz Ashraf [a], Eugen Fernandez [b]

[a] Department of Electrical Engineering, Aligarh Muslim University, Aligarh, Uttar Pardesh 202002, India
[b] Department of Electrical Engineering, Indian Institute of Technology, Roorkee, Uttarakhand 247667, India

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ABSTRACT

Renewable energy is widely used in the cost-effective electrification of rural communities. This article discusses a better electricity option for the residents of Uttarakhand's (India) rural areas. The proposed technology is utilized to create a standalone microgrid that reliably fulfills the area's energy demands at a low cost of energy. The suggested system focuses on the hybrid energy model's optimal sizing and sensitivity analysis. The integrated model uses locally accessible energy resources such as solar photovoltaic, micro-hydropower, biogas, battery, biomass, and wind energy systems to meet the area's overall electrical energy requirement. In the simulation, the proposed energy framework considers the intermittent nature of renewable resources. The proposed microgrid model is optimized by utilizing differential evolution (DE) to reduce the overall cost of energy and sizing of the system. The effectiveness of DE is tested for optimization problems, and its results are compared against particle swarm optimization and genetic algorithms. Sensitivity analysis of the cost of energy and the total net present cost demonstrates the impact of modifying the input parameters. The simulation findings reveal that DE optimizes the system compared to its contemporaries, PSO and GA. According to the findings, the optimal configuration has a total net present cost of $ 7,12,532.00, and a cost of energy is 0.14$/kWh. The suggested model for the electrification of the rural community is quite effective in optimizing the system's cost and energy, and sizing.

1. Introduction

Fossil fuels have generated a sizable portion of power production worldwide. These fuels emit greenhouse gases into the environment. Attempts are being made to reduce greenhouse gas emissions. Energy from renewable sources is the ideal alternative to conventional fuels. Enormous availability and a clean-green environment-friendly resource are the primary concern for an option. The use of renewable energy resources has risen significantly (Fatin Ishraque et al., 2021). The IEA suggested that 30% of total energy use be satisfied by using renewable energy resources by 2022(*IEA*, 2021.). Several renewable resources, including tidal, SPV, BM, WES, BG, and MHP, are frequently used to meet the energy needs of both rural and urban areas. The solar photovoltaic energy system is a clean and environmentally friendly form of energy that converts solar energy to electrical energy. The electricity shares from wind energy systems increased dramatically. It is expected that by 2050 wind energy share will be 15-18% (Ferraz de Andrade Santos et al., 2020). Hydro energy systems are supposed to be a clean energy resource. It is the world's most significant source of electricity, with an installed capacity of 1,308GW (IHA, 2021). Biomass and biogas are other adequate sources of generation of electricity. It contributes much electricity to rural electrification (Ellabban & Alassi, 2019).

Due to overpopulation, industrialization, and urbanization, a sustainable energy system is strongly linked to a country's economic development. To meet the energy demands, developing economies rely on fossil fuels. Renewable energy technology can help any country's economy grow. Therefore, due to the increased traditional fuel costs and the issues that energy security creates due to lack of energy, it is critical to pursue renewable energy alternatives. However, renewable energy solutions must be pursued due to the rising diminution of conventional fuels and the increasing environmental GHG emissions. Renewable energy sources are becoming increasingly popular in meeting energy demands worldwide (Kamal & Ashraf, 2021a). Microgrid systems based on renewable energy are the most important means to provide electricity to remote populations without disrupting service. Research on the study and evolution of renewable energy integration for power supply in remote rural locations has been published in various literature.

Renewable energy resources (RER), such as WES, PV, MHP, BM, and BG, are used to suggest a microgrid for the electrification of residents of

* Corresponding author.
E-mail addresses: mustafakamalece@gmail.com (Md.M. Kamal), iashraf@rediffmail.com (I. Ashraf), eugenefdz@rediffmail.com (E. Fernandez).

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Nomenclature

List of abbreviations

ABC	artificial bee colony
BAT	battery
BG	biogas
BBO	bibliography based optimization
BGES	biogas energy system
BM	biomass
COE	cost of energy
DE	differential evolution
DOD	depth of discharge
DG	diesel generator
DPSP	deficiency of power supply probability
DESPD	Directorate of Economics & Statistics, Planning Department
FA	firefly algorithm
GA	genetic algorithm
HOMER	hybrid optimization of multiple energy resources
IEA	international energy agency
IHA	international hydro association
IRES	integrated renewable energy system
LPSP	loss of power supply probability
MHP	micro-hydro power
MILP	mixed integer liner programming
MPSO	multi-objective particle swarm optimization
NASA	National Aeronautics and Space Administration
O&M	operation and maintenance
PSO	particle swarm optimization
PS	pattern search
RER	renewable energy resource
SPV	solar photovoltaic
SOC	state of charge
TNPC	total net present cost
UPCL	Uttarakhand power corporation limited
UREDA	Uttarakhand renewable energy development agency
WES	wind energy system
WT	wind turbine

List of symbols

A	area(m^2)
A_{Tbio}	availability of biomass(tons/yr.)
E_t(t)	solar radiation (kW/m^2)
E_s	standard solar radiation (1 kW/m^2)
I_c	short circuit current (A)
I_{PV}	PV output current(A)
$I_{\propto}$	generated current (A)
I_{Pm}	photo diode current(a)
I_s	reverse saturation current(A)
p_r	rated power(kW)
P_W	power output of wind turbine (kW)
Q(t)	discharge rate(m^3/s)
Q_{BG}	availability of biogas(tons/yr.)
$T_a(t)$	ambient temperature (K)
T_h	pperating hour per day(day/hr.)
T_{ref}	reference temperature (K)
V_{PV}	module output voltage(V)
V_{MP}	module voltage at maximum power(V)
ρ	water density (kg/m^3)
α	open circuit temperature co-efficient(/°K)
i	running index of population
β	short circuit temperature coefficient(/°K)
Δt	step length
$	US dollar

urban and rural communities. Various RER intergraded to form an efficient resident electrification system (Suman et al., 2021). The hybrid microgrid system is an efficient, dependable, resilient, environmentally friendly, and cost-effective way of electricity. It can be used as an on-grid or standalone system (Kumar et al., 2021). SPV and WES's inconsistency and weather dependence provide technological and economic hurdles to their planning and optimal design(Bosisio et al., 2019). The complexity of the model is affected by the application and load connected, and a micro-grid should be built to handle the required load even though one of the components is inconsistent due to weather conditions (Ribó-Pérez et al., 2021). As a result, resource selection is influenced by the environment of the area and storage units. In developing countries like India, electrification using biomass and biogas is cost-effective (Satsangi et al., 2019).

The planning and optimization of microgrids include a variety of technologies and methods. The HOMER optimization tool for a microgrid model for the residential load of the province of Egypt has been suggested for the electrification of rural communities. The proposed algorithm combined PV/Wind/biomass to electrify the rural population and optimized the energy model for the least per unit energy cost (Gamil et al., 2021). The metaheuristic technique is widely used in minimizing electricity costs. For rural electrification combining hybrid energy resources is proposed by Balderrama et al. (2019). They proposed a realistic and economic power resolution for rural electrification of Bolivia in the absence of grid connectivity. Similar studies were carried out for rural electrification in the hilly region of Indian villages. The ideal scale of a hybrid energy system was proposed to meet the daily electricity needs of the area. For the remote Indian village, a hybrid energy model was utilized to minimize the overall cost of energy for a group of villages in Uttarakhand (India). MHP/SPV/ BGS/ WES are locally available energy resources that have been integrated to satisfy the community's energy demand. For the reliable supply to the area, LSPS is the indices (Chauhan & Saini, 2016b).

Various optimization approaches are commonly employed in the hybrid energy system to model and component sizing rural microgrids. There are a variety of tools available for the planning and cost minimization of the microgrid. For this instance Zhao et al. (2021) suggested a novel method for the size and cost optimization of the rural microgrid. They incorporate renewable fractions, GHG emissions from a DG, TNPC, and overall operating cost while planning the integrated model. Three alternative intelligent algorithms, pattern search algorithm-frog leaping, were utilized to optimize energy sizing and investigate comparative results. PSO and GA also validated the effectiveness of the suggested model results. Authors (Sanajaoba, 2019) created a microgrid for the rural population in Manipur(India). The system consists of PV/WT/MHP/BAT and DG. The formulated microgrid model is optimized for the least cost and sizing by GA, PSO, and FA. In a similar study (Nurunnabi et al., 2019), the authors proposed a microgrid model for a rural location in Bangladesh to analyze the optimal configuration of the microgrid HOMER Pro software.

The sensitivity and the impact of greenhouse gases were studied. Finally, a detailed feasibility chart is prepared and suggests the low-cost model for electrification. The system is made up of PV, WT, DG, and BAT. Authors (Mosa & Ali, 2021) For the reduction of electricity costs and components' size, a more advanced MILP approach was applied. They proposed a microgrid for a residential neighborhood in Egypt's northeast. The suggested model's economic and environmental effects were assessed using the optimization method. Findings suggest that batteries reduce the generation and environmental costs and compensate for the generation shortage. The authors (Sanajaoba Singh &

Fernandez, 2018) suggest that because of the geographical outline of Uttarakhand state, the state has enormous potential for several RER. The local energy needs can be met by adopting the right combination of resources. Renewable resources can be used to provide energy and can form smaller microgrid models. They used the cuckoo algorithm for sizing and optimal cost suggestion and compared results with GA optimization. Rural electrification promotes the societal and financial well-being of the people who live there while also hastening the nation's progress. As a result, increasing the power infrastructure, whether grid-connected or standalone, can help increase rural electrification. Grid connectivity has reached rural areas amid government efforts, but only 55.33% of India's rural population has access to electricity (Pelz et al., 2021). In rural locations, a significant portion of the community remains un-electrified or has unreliable electrification sources (Yadav et al., 2019). Because grid connectivity is not possible in hilly places due to the topographical structure, the microgrid is the only alternative for providing energy access in that area (Singh & Basak, 2021). Small microgrids and nano grids can be built to provide clean electricity to remote locations. Renewable energy sources such as SPV, WES, MHP, BG, and BM, which are abundantly available, can be used to meet local energy demands. These energy resources can be combined to produce a hybrid energy system (Aklin et al., 2018).

Numerous software packages are utilized in the planning and development of the microgrid, including HOMER pro. The authors (Das et al., 2021) presented microgrids' feasibility and sensitivity analysis for several rural locations in Bangladesh. The suggested model takes into account different renewable resource combinations while modelling. The optimization was done with the help of the HOMER Pro program. In Karnataka-based Indian communities, a similar study was conducted. They built an energy model using the optimum set of a PV/DG/BAT and WT to electrify the rural population of Indian villages to minimize total energy costs. The metaheuristics algorithm PSO was utilized for the cost of energy optimization, sizing, and sensitivity analysis (Ramesh & Saini, 2020). The authors (Han et al., 2018) proposed a hybrid energy approach for rural electrification. They designed a hybrid model by integrating an SPV with a BAT. To reduce the overall cost of the installation, they assessed the total cost of operation and used a GA optimization algorithm to optimize the COE. They developed a framework based on power reliability to calculate the probability of an outage. The authors (Robert & Gopalan, 2018) suggested a microgrid for the three locations of the state of Kerala (India). The system consists of PV/WT/BAT and DG. HOMER Pro software was used to optimize the component sizing and energy cost. The finding suggests that COE is 0.17 ($/kWh), meeting 93% of the total load. In another study (Kharrich et al., 2021), a suggested hybrid model was proposed for rural electrification in Morocco. The formulated objective was optimized by using the MPSO algorithm and six sigma tools. The algorithm calculates the NPC, COE, and greenhouse gas emissions. Aside from effective electrification, sensitivity analysis with parameter change was also carried out. A similar study carried out by Mandal et al. (2018). They proposed microgrid model for Bangladesh's rural community. The hybrid system is designed to meet the daily load demand of 242.56kWh for a remote, isolated village community, with a peak load requirement of 51.52kW. With a 57kW DG, 73kW PV arrays, 28kW inverters, and a 387kWh nominal capacity battery bank, the best hybrid system has a minimum COE of 0.37$/kWh and a TNPC of $3,57,284. HOMER Pro simulation software was utilized for sizing and sensitivity analysis. In another study the optimization algorithm used for planning of microgrid. The goal of the planning process is to provide consistent electricity delivery while keeping deployment and operations expenses to a minimum. The planning problem is expressed as an objective which minimized using MILP. Due to the problem's complexity, the suggested approach uses a two-stage framework to produce an efficient planning solution. The results are optimized using GA and PSO, and the two optimization strategies are compared (Mohamed et al., 2019). The authors (Upadhyay & Sharma, 2016) devised a methodology for sizing that takes into



Fig. 1. Hybrid microgrid system.

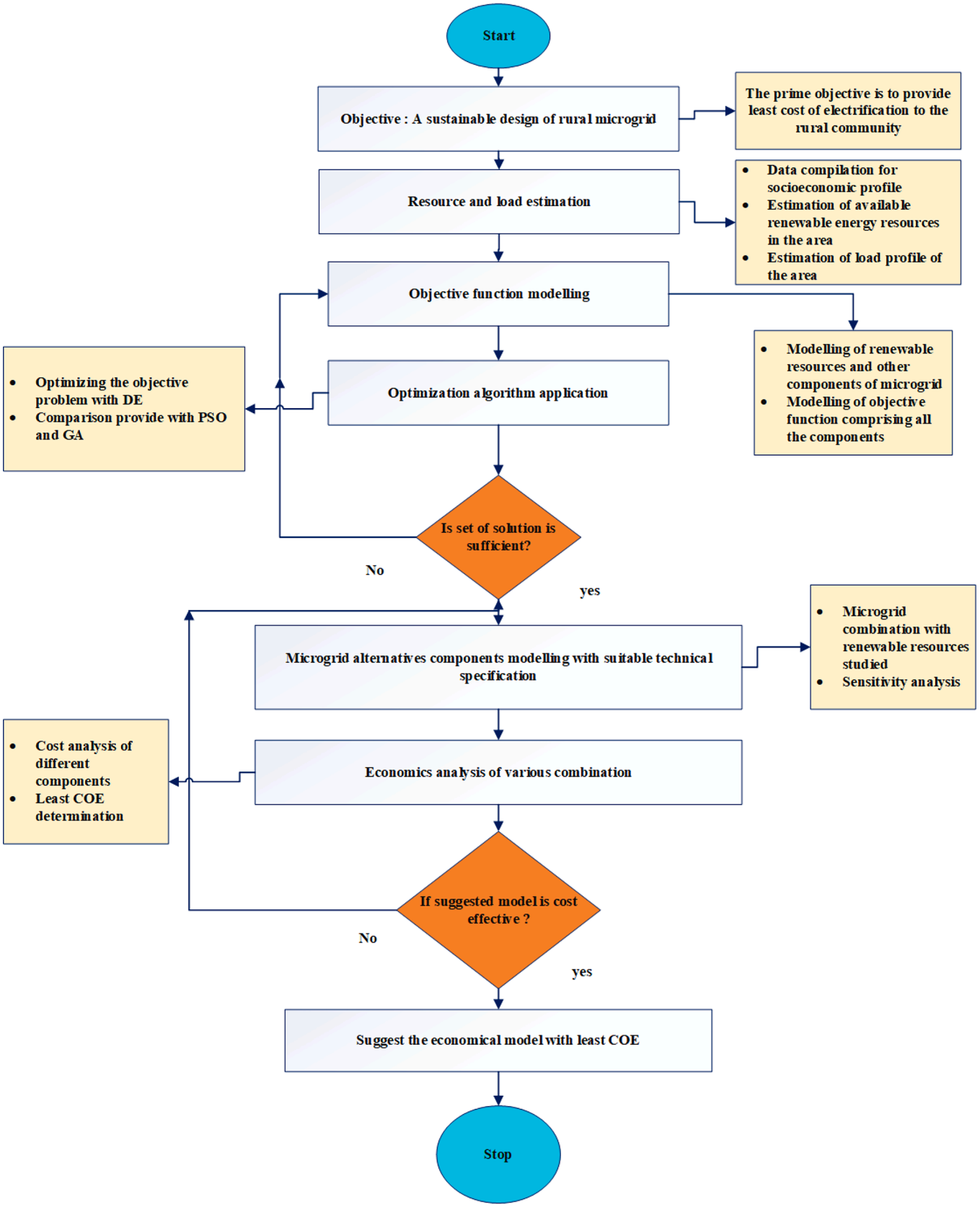


Fig. 2. Methodology of the proposed work.

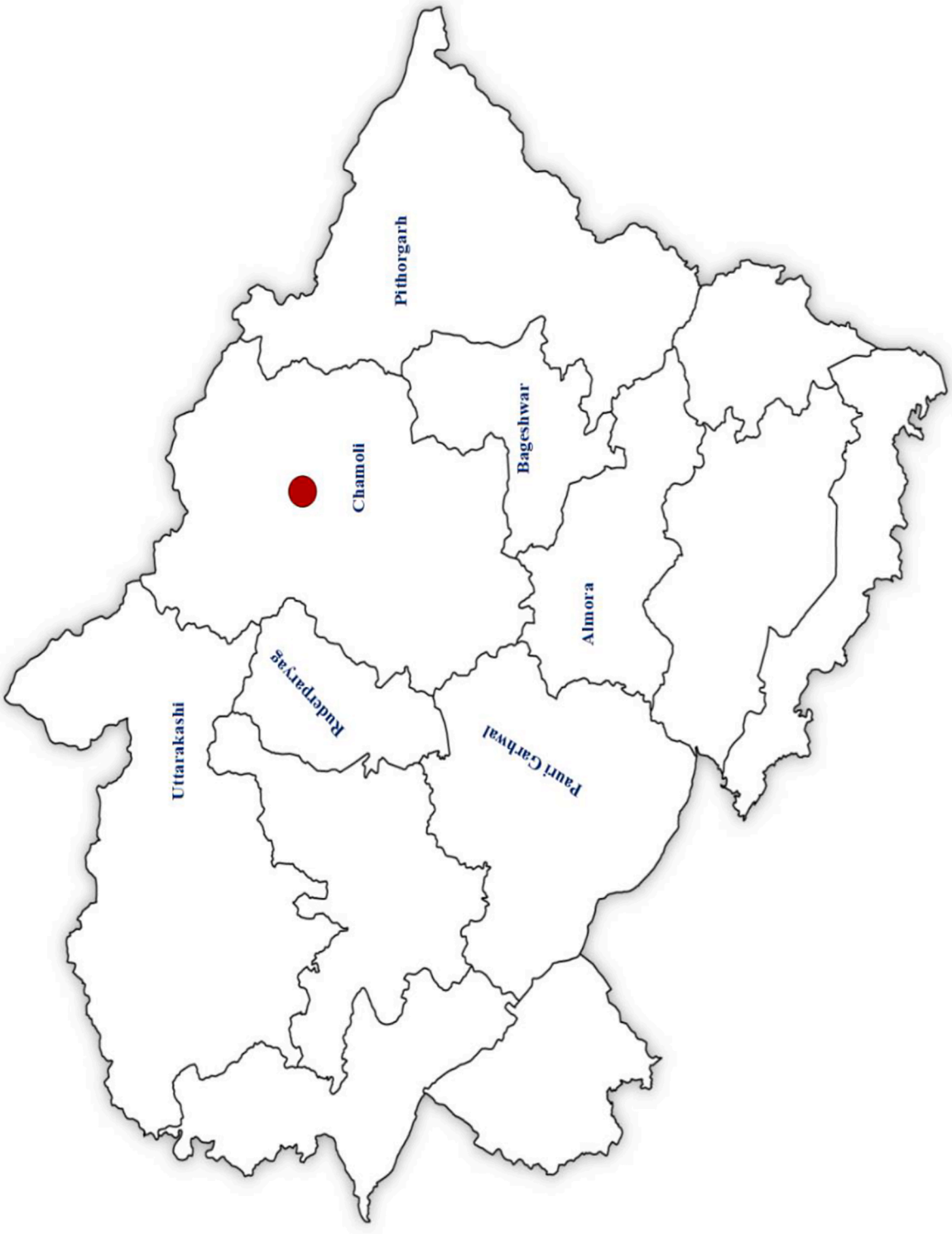


Fig. 3. Location of the study area.

account various energy resources. They optimized with BBO and compared the outcomes to PSO and GA. PSO and GA have limitations that can be avoided by modeling and optimizing the objective function in the size problem using the BBO technique. Authors (Dhaliwal et al., 2021) presenting effective microgrid management using various forms of renewable energy resources. Uncertainties in production are also taken into account while modeling the system. The COE is optimized by a new optimization method based on the bat algorithm.

1.1. The main contribution of the study

Several studies on the rural microgrid have been conducted, each using a unique strategy to investigate the topic and achieve specific goals, as stated in the preceding section. Most of them view the functioning of a rural microgrid with a high penetration of renewable energy sources as a crucial step in avoiding a traditional fossil fuel-based system and moving toward a power grid with a high proportion of carbon-free energy production units. The vast majority have looked at the benefits of this shift from numerous angles, including economic and environmental, and depending on the issue type, and they have conducted varied evaluations. The idea of modelling RESs' stochastic behavior has been proposed as a remedy. Based on previously published research, this paper offers a multi-objective optimal sizing analysis of renewable energy microgrids for residential areas. The main goals of the suggested approaches are to keep overhead costs to a minimum and offer residents electricity at the lowest possible price. Evaluating the profit opportunity, dependability and cost minimization, and CO_2 emissions. The following list includes the study's main contributions:

i The techno-economic analysis is provided for optimal scheduling of the rural microgrid equipped with renewable energy generating systems while considering technical and economic goals.
ii The electrification of the community provides by integrating available energy resources in the vicinity. For the case study, a group of villages in the hilly area of Uttarakhand (India) has been taken.
iii Microgrids are formed by integrating renewable energy sources like PV, WES, BM, BG, and MHP. The local social and economic activity can be improved by providing reliable electrification.
iv Demonstrating a multi-objective optimal sizing strategy for electrifying rural communities offers an affordable electrification alternative.
v For cost optimization, a differential evolution algorithm-based technique is being used. PSO and GA comparisons are made with the results.
vi The most affordable method of electrification is investigated using various renewable resource combinations.
vii Sensitivity analysis is carried out for the best case by modifying the most sensitive factors, which substantially impact the system's economy.

The remaining sections of the paper are organized as follows: section:1 provides background theory and literature support; section: 2 describes the study's methodology; section: 3 describes the load modelling and area description; section: 4 provides the mathematical modeling of the microgrid components; section: 5 provides the problem formulation and description of related constraints; section: 6 explain the case study of the selected location; section: 7 describes the results and discussion; seticon:8 provides the sensitivity analysis; The comparison of the current investigation with the prior study is described in part 9; the conclusion is presented in Section 10.

2. Methodology

The localized renewable resources constitute the microgrid model. Fig. 1 depicts a schematic of the microgrid. Fig. 2 depicts the approach taken by the integrated model to utilize available energy resources and reduce overall energy costs. The literature reports a variety of strategies, including GA (Fan et al., 2022), harmony search algorithm (Chauhan & Saini, 2016a), GWO (Dey et al., 2022), PSO (Suman et al., 2021), cuckoo algorithm (Sanajaoba & Fernandez, 2016), DE (Kamal et al., 2022), Jaya algorithm (Alshammari & Asumadu, 2020) etc.

Each of these algorithms can handle the nonlinear variations in renewable energy system components. The model is developed by integrating available resources and optimizing the output using different algorithms added to the viability assessments carried out by earlier versions. The purpose of this study is to electrify rural populations with substantial renewable resources.

The available resources MHP, SPV, BM, BG and WT are used to model the system. for the backup and proving power for the community storage units are also used. The microgrid can be formed from the available resources and several combinations, which can be analyzed for feasibility and the least cost of operation. The model first compiles all the potential sets from the inputs it receives from the various sources. But not all prospective sets will have the right energy and economic properties. To select a batter combination, the first remove combinations of poorly organized sources, particularly regarding variability and flexibility aspects; the model then adopts a qualitative approach.

Due to their inherent inability to provide significant production flexibility, the model also disallows combinations containing more than two low-capacity sources. The model automatically disqualifies combinations involving more than one plant using the same renewable resource, as such combinations would inherently reduce the system's overall flexibility and frequently lead to the accumulation of the same type of variability. However, due to the seasonality of the resources, only four cases are considered for the current study. The following scenarios are looked into to see which electrification option is the most affordable and feasible.

Case 1: MHP/BG/BM/BAT
Case 2: MHP/BG/BM/WT/BAT
Case 3: MHP/BG/BM/SPV/BAT
Case 4: MHP/BG/BM/SPV/WT/BAT

The fundamental purpose of this research is to provide a feasible, long-term energy system for electrifying a self-sufficient rural hamlet. Renewable energy sources and energy storage units are used in the system. The suggested method is aimed at the inhabitants of Uttarakhand, India's rural areas.

3. Area selection and load assessment

The current case study is used to verify the projected microgrid technology for the least expensive electrification in the region. The investigated area is situated in the Chamoli district of Uttarakhand (India). The un-electrified villages of the region are selected for the case study. The study involves the selection of a region, simulation, and validating of the results.

3.1. Study area

The hilly state is one of the wealthiest state zones regarding renewable energy resources. The region contains a sizable Himalayan Forest, 300 days of sunshine, and several waterfalls (Kamal & Ashraf, 2021b). The state currently pursues solar energy with a more than 550 MW capacity and a hydropower capacity of 1550 MW.

There are 147 MW of capacity over 18 minor and 33 mini-hydro plants. Several such hydropower projects are being developed, with a combined installed capacity of 9.74 MW(*UREDA*,2021.). These projects are connected to the grid(*UPCL*,2022.). There are groups of villages in the study area investigated. Geographically, the region is situated at 30.42°N and 79.33°E. The area's location on the state map is shown in Fig. 3 (*Chamoli District 2022*).

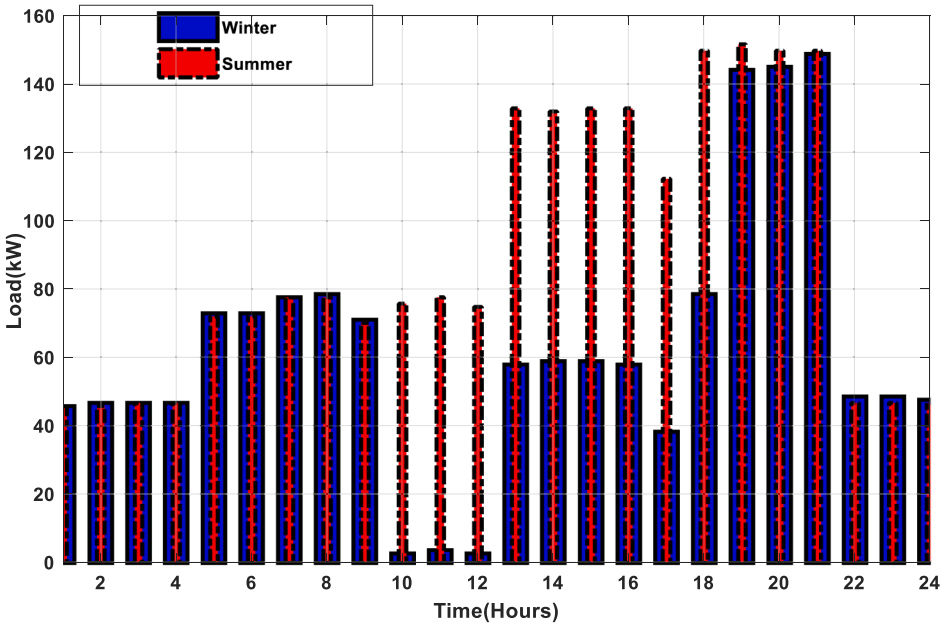


Fig. 4. Hourly demand of two seasons.

Table 1
Organization and classification of types of loads.

Total load									
Domestics loads					Community loads				
Apparatus	Types	Ratings(W)	No.	Total(W)	Apparatus	Types	Ratings(W)	No.	Total(W)
Homes	CFL	15	1120	16800	GPO	Fans	65	09	585
	Television	75	600	45000		CFL	35	9	540
	Fans	65	600	38000		Computer	100	5	500
	Radios	15	600	9000		Printer	40	3	120
Commercial loads					Province	Lights	35	5	150
						Heaters	1500	5	7500
						Fans	65	5	325
						coolers	250	5	1250
					Hospitals	Lights	35	15	525
Apparatus	Types	Ratings(W)	No.	Total(W)		Fans	65	15	975
shops	Lights	30	30	900		Heater	1500	5	7500
	Fans	35	25	875		Refrigerator	475	5	2375
Flour mills		5000	6	3000		Cooler	250	5	1250
Industrial Loads					Schools	Lights	30	35	1050
Sawmill		5000	5	25000		Fans	60	70	4200
					Arc lamps	Lights	70	30	2100
					Pumps		746	35	26110

3.2. Load assessment of the study area

The study areas are considered to be isolated rural communities. Fig. 3 suggests the location of the study area. The area is a hilly region and is surrounded by the district, which is located in the Western Himalayas, is bordered by Tehri Garhwal in the west, Almora in the south, Rudraprayag in the south-west, Pithoragarh in the south-west, and Uttarkashi in the north-west. The region is surrounded by the Himalayan River Alakhnanda and Ramganga (Singh & Chatterjee, 2022). So, these geographical locations and lack of proper transport make them isolated in the hilly region. As a result, only a small amount of electricity is used for the daily energy requirement.

Most appliances and equipment are used in the evening, which results in significant energy consumption; only agriculture adds to the day load profile. The agricultural load includes the need for energy for a water-lifting pump and a machine for threshing crops and cutting fodder. The area's projected load is assessed to provide a reliable electricity supply. The electrical load and energy consumed for cooking were evaluated in the research region. The area load was modelled as per the survey, past reports(DESPD,2021.), and the literature available for the concerned area (Bhatt et al., 2016; Chauhan & Saini, 2016c). The available BM and BG supplies can meet the cooking energy needs. There is much forestry in the research site, and people have many cattle, both of which can support biogas, biomass plants, and agriculture. Using animal manure and forest trash can satisfy the energy requirements for cooking. By studying and estimating the per capita biogas use, the daily requirement for cooking energy is 800 m^3/day. Biogas usage is estimated to be 0.46 m^3/day per person (Kumar & Saini, 2020).

The entire year is subdivided into the summer season (April-September), and the winter season (October-March) to analyze the overall electrical energy consumption of the research region. The area's total load is approximated. The load modelling of the area divided into commercial, domestic, community, and agricultural energy usage is divided into four categories. Fig. 4 shows the entire load of two seasons. Table 1 suggests the loads' classification and rating (Chaurasia et al., 2021). In summer and winter, the peak energy demand is 155.78 kW and

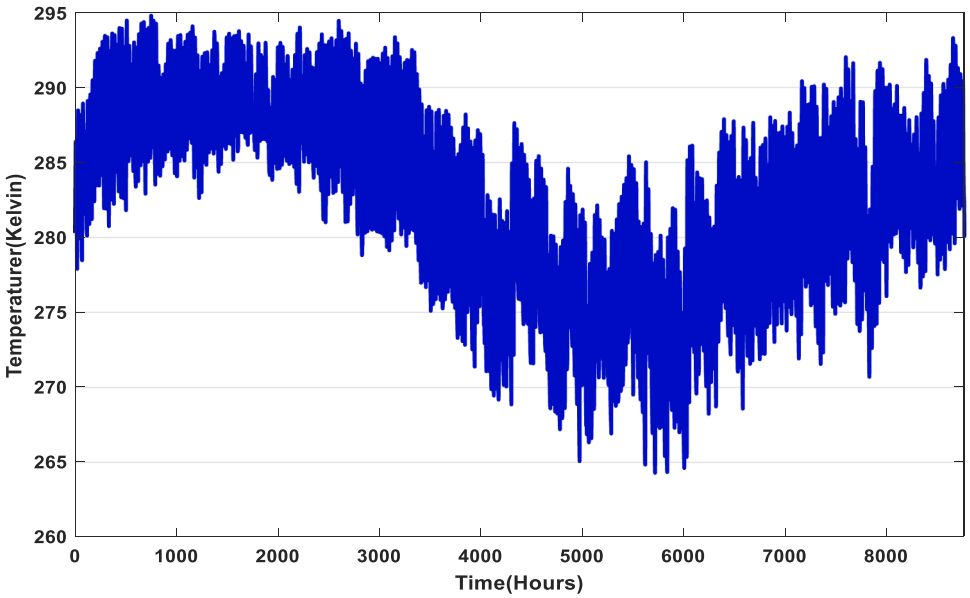


Fig. 5. Temperature variation in the study area.

149.77 kW, as shown in Fig. 4. The peak load of the area is reported in late night hours as regional small-scale industries are working. The summer peak load is reported in the 18th hour, and the winter peak load is reported in the 21st hour of operation. The research area's monthly load distribution is shown in Fig. 4.

4. Mathematical modeling of resources

An extensive survey was conducted to estimate the available resources in the research area, including quick visits and investigations into government data. Statistics on solar radiation, biomass, biogas, wind energy, animal waste, and water outflow were examined. The metrological data used in microgrid modeling is directly taken from NASA metrological data sources (*NASA*, 2021.). This section describes the mathematical modeling and availability of resources in the study area.

4.1. Solar photovoltaic system

The SPV system consists of several PV modules. For modelling the PV system, the single-diode model of the PV module is considered. The voltage output of the diode model is represented by Eq. (1) (Mishra et al., 2020)

$$V_{PV} = V_{MP}\left[1 + 0.0539\log\left(\frac{E_t}{E_s}\right)\right] + \alpha(T_a(t))) + 0.02E_t(\mathrm{t}) \tag{1}$$

The PV module current (I_{Pm}) depends on the incident solar radiation, and temperature can be calculated by Eq. (2) (Chen, 2013).

$$I_{Pm} = [I_c + \beta T_a(t) - T_r] \times \frac{E_t}{E_s}] \tag{2}$$

The reverse saturation current depends on the solar radiation, and temperature can be represented by Eq. (3) (Kharrich et al., 2021):

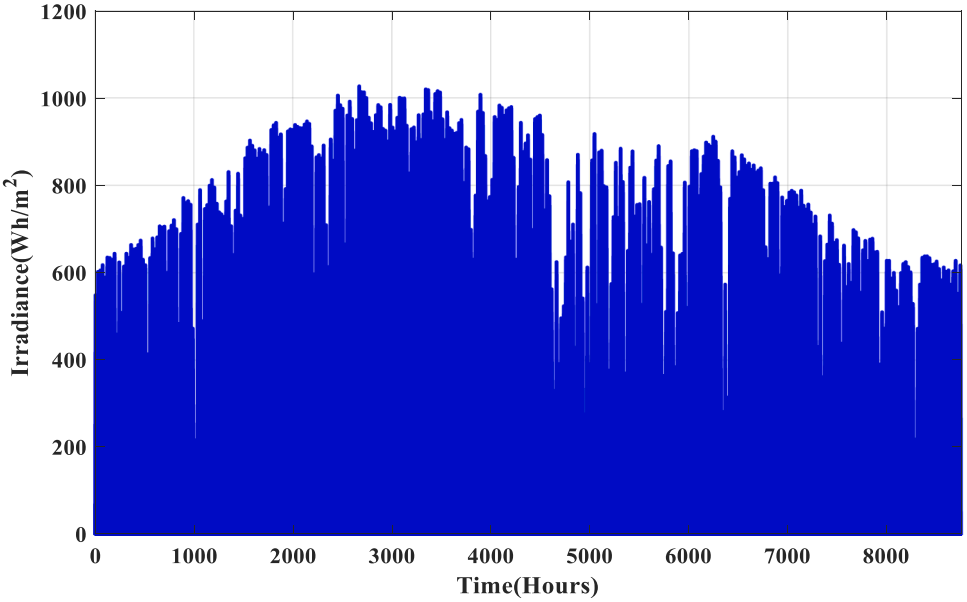


Fig. 6. Yearly irradiance variation of the study area.

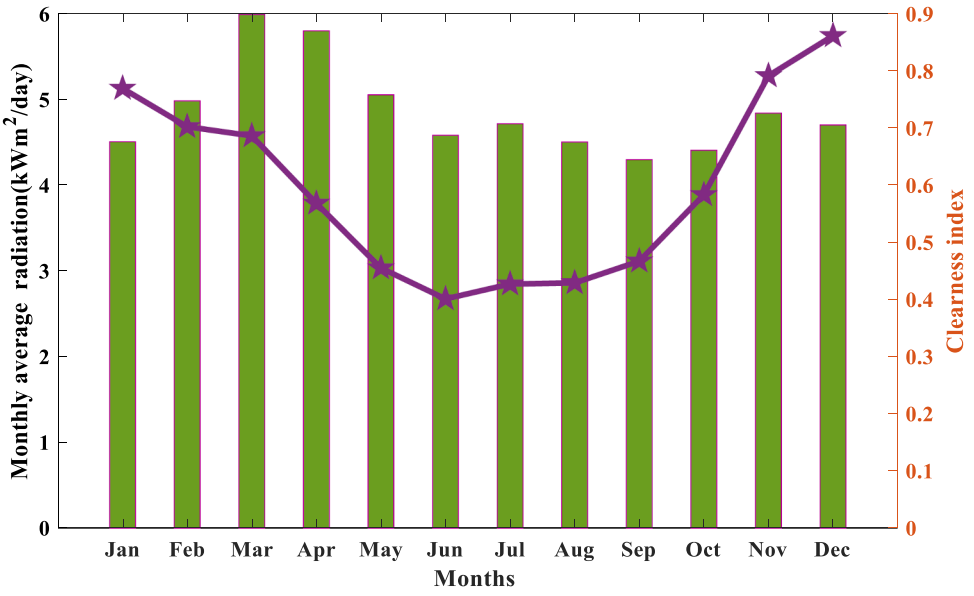


Fig. 7. Monthly average solar radiation and clearness-index.

$$I_s(t) = I_{or}\left(\frac{T(t)}{T_{ref}}\right)^3 \times exp\left(\frac{qE_g(1/T_r - 1/T(t))}{KT(t)}\right) \quad (3)$$

Where, I_s is the reverse saturation current at the given reference temperature T_{ref}, E_g is the bandgap, q is the charge of an electron.

The photovoltaic cell is modeled by the nonlinear current connected to the series with the resistance in the series. The output current (I_{PV}) depends on the ambient temperature and solar irradiation represented by Eq. (4) (Sanajaoba Singh & Fernandez, 2018)

$$I_{PV}(t) = I_{\propto}(t) - I_s(t)\left[exp^{(q(V_{\propto}(t)+I_{\propto}(t)R_s/A_cKT(t))} - 1\right] \quad (4)$$

Where $I_{\propto}$ is the generated current under given irradiance, I_s is the reverse saturation current, T is the cell temperature, K is the Boltzmann constant, R_s is the cell-intrinsic resistance, A_c is the ideal cell deviation from ideal PN junction characteristics.

At any hour t, power output(P_{PV}) of SPV system is estimated by Eq. (5) (Upadhyay & Sharma, 2016)

$$P_{PV} = N_{PV}V_{PV}I_{PV}(t)/1000 \quad (5)$$

Energy generated by the solar PV system (E_{PV}) estimated by the Eq. (6).

$$E_{PV} = P_{PV}\Delta t \quad (6)$$

Several PV modules combined formed the SPV system. Solar radiation impacting the SPV system's surface is transformed into electricity. Solar radiation data was collected from 30.15 latitudes to 79.14 longitudes. The solar data is derived from the NASA solar data viewer. The region's mean annual solar radiation is 5.32 kWh/m^2, with the most significant solar radiation recorded in March at 6.0 kWh/m^2(*NASA*, 2021.). Because the location has more than 300 solar days, the potential for solar energy harassment is very significant (Chauhan & Saini, 2015). The research region's overall yearly solar energy potential has been calculated to be 22,369 kWh/m^2/yr (Mishra et al., 2020). The area's annual solar radiation and temperature variation are depicted in Figs. 5–7.

4.2. Micro-hydro power system

The state of Uttarakhand has excellent hydroelectric potential because of the Himalayan area. The Himalayas are home to numerous rivers and waterfalls. Several hydro and run-of-the-river (ROR) projects are running in the district. The micro-hydro projects at Rishiganga on the Rishiganga river and TapovanVishnugad on the Alaknanda River. The National Thermal Power Corporation Limited (NTPC) barrage at Tapovan town is one of the RORs in the Dhauliganga river that has been affected. Other ROR projects in the downstream region, such as the Vishnugad Pipalkoti and Srinagar hydroelectric power plants, Raini hamlet, located at the confluence of the Dhauliganga and Rishiganga rivers (Siddque et al., 2022). The proposed location's industrial, commercial, and domestic power needs can be met by hydropower in hilly locations. The hydro turbine uses the mechanical energy that is created by the water pressure to produce electricity. The micro-hydro power plant's hourly output is based on the net head and discharge available at the specific location. The relationship between pressure and volume flow rate determines the available power. This can be used to calculate the Eq. (7) (Ma et al., 2015).

$$P_{MHP}(t) = \rho \times Q(t) \times g \times H \times \eta_{MHG} \quad (7)$$

Where, H is the difference in elevation between the highest and outlet points, ρ is water density, Q(t) is the rate of discharge (m^3/s).

4.3. Biomass Energy System

Biomass is converted into potential electricity using biomass generators. The biomass energy system in the research area primarily consists of agricultural waste and forest debris. The Himalayan Forest surrounds the location. The Western Himalayas, located in India's Uttarakhand state, covers a total area of 53,484 square kilometers, with a total forest cover of 24,303 square kilometers (Singh & Chatterjee, 2022). Because of their high species variety and high net primary output, tropical forests fix more biomass and serve as a significant potential carbon sink (Joshi et al., 2021). So the overall biomass potential of the region is 92.21Tons/yr. from agriculture waste and 36.36 Tons/yr. from forest foliage (Rajanna & Saini, 2016). Systems based on biomass gasifiers are used to generate power from biomass. The biomass gasifier operated for 10hrs/day, and the power capacity of the gasifier is 30 kW used in the present microgrid planning. The mathematical power of biomass is represented by Eq. (8) (Upadhyay & Sharma, 2016).

$$P_{BMG}(t) = \frac{(A_{Tbio} \times CV_{BM} \times \eta_{BMG})}{365 \times 8760 \times \partial} \quad (8)$$

Where, A_{Tbio} is the availability of biomass, η_{BMG} is the overall efficiency

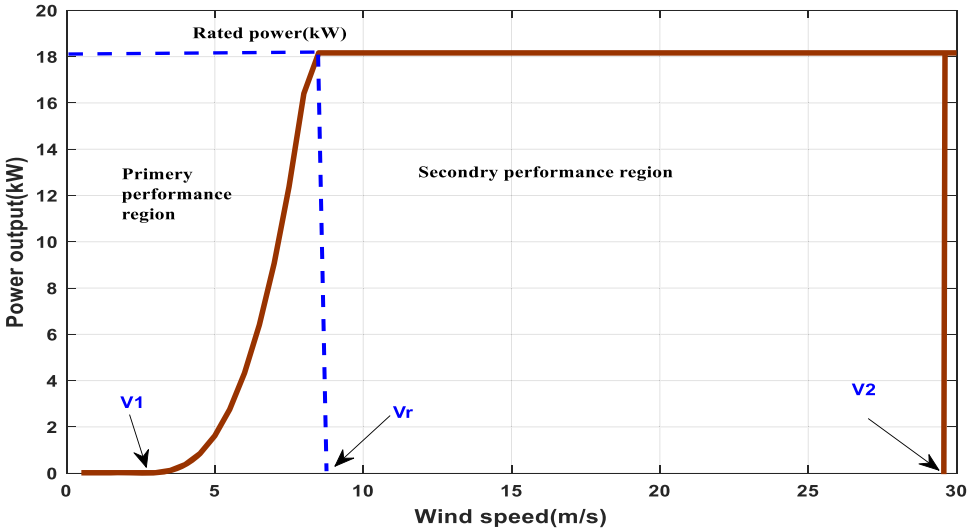


Fig. 8. Power curve of wind turbine.

of biomass power plant CV_{BM} is the calorific value of biomass. ∂ is the operating hours per day.

Dump waste is transformed into electricity using a biogas generator. For the most part, biogas is converted using animal manure. A detailed door survey is conducted, including counting the number of animals and considering potential storage of wood and forest foliage, to determine the study's overall biogas potential (Choudhary et al., 2020). For counting the animals and possible storage of foliage, a few questionnaires were asked to the resident. Based on the resident response and data published by the government potential of biogas is estimated (Statical report 2021.). The total population of livestock by biogas can generate estimated. The study area has many Cattel's, which provide biogas material to the gasifier. Dung is available at the site at a rate of 18,104 Kg /day. The manure produces 650.50 m^3/day of biogas, of which 401.58 m^3/day is used for cooking, and the rest is used for power generation (Chauhan & Saini, 2016b). The biogas generator works nine hrs. a day, and the total output power is 30 kW. The biogas availability is calculated by eq. (9) (Singh & Basak, 2021).

$$P_{BCG}\ (t) = [Q_{BG} \times CV_{BG} \times \eta_{BCG})/\ 860\ \times T_h] \tag{9}$$

Where, Q_{BG} is the biogas availability, CV_{BG} is the calorific value of biogas, η_{BCG} is the overall efficiency of biogas gasifier engine, T_h is the working hour per day.

It is assumed that the quantity of dung is 1.5 kg/day dung from goats, 11kg/day of dung from mule/cow/horse, and 16 kg/day from buffalo.

4.4. Wind energy system

The mechanical power of the wind is transformed into electrical energy by a WT. The velocity of air is transferred into rotational motion by a turbine. The wind flows or spins the rotors as it passes through a wind turbine. Wind speed at a set hub height and the speed parameters of the turbine determine the power output of a wind energy generator at a given location. The power-law equation can be used to compute the

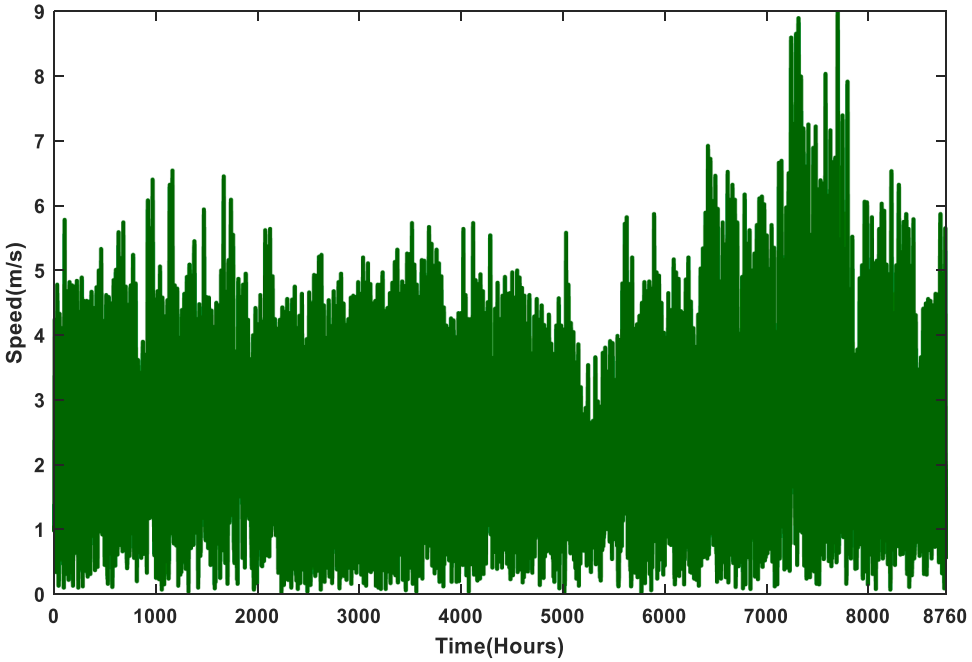


Fig. 9. Annual speed variation of wind in the study area.

wind speed at a height represented by Eq. (10) (Suman et al., 2021).

$$(v / v_r) = (h/h_r)^n \tag{10}$$

Where v and v_r is the wind speed at the hub and reference height h and h_r, and n is the power-law exponent. power ($P_{WES}(t)$) generation varies with wind velocity(v) for the typical wind turbine characteristics. The area under the curve from a wind turbine's cut-in speed and cut-out determines how much power it can produce. The wind speed before the cut-in speed (v_c) and cut-out (v_o) cut-in speed and after the cut-out speed are not used to produce electricity. In light of this, the power density can be estimated by Eq. (11) (kamal et al., 2021).

$$P_{WES}(t) = \begin{cases} 0 & v\langle v_{ci},\ v\rangle v_{co} \\ \dfrac{p_r}{\left(V_r^3 - v_c^3\right)} v^3 - \dfrac{v_c^s}{\left(V_r^3 - v_c^3\right)} p_r & v_{ci} \le v < v_r \\ 0 & v_r \le v < v_{c0} \end{cases} \tag{11}$$

The rated power of wind turbine expressed by the Eq. (12) (Singh et al., 2016)

$$P_r = \frac{1}{2} \times \rho_W \times A_W \times c_p \times v_r^3 \tag{12}$$

Where P_r is the rated power of wind turbine (W), A_W (m^2), c_p is the maximum power co-efficient, ρ_W is the air density ($Kg\ /m^3$).

Fig. 8 shows the characteristics of the used WT and the variation of wind speed in the area. In India, wind turbines are designed for wind speeds of 10m/s and above but for small-scale applications. For such wind turbines, Uttarakhand (India) has some favorable locations (Chauhan & Saini, 2016a). The mean annual wind speed of the area is 5.11 m/s. Fig. 9 shows the annual wind speed of the area (*NASA*, 2021.). The present study used the BS-HV4-10K model wind turbine.

4.5. Modeling of the battery system

Renewable energy sources are only usable for a finite amount of time. As a result, energy supply reliability cannot be guaranteed without backup energy sources. The microgrid system includes a battery storage system to store excess energy and deliver electricity when renewable energy generation is insufficient to satisfy demand. The following formula can be used to compute the amount of energy stored in the battery bank (Hossain et al., 2017a).

$$E_b(t) = E_b(\mathrm{t}-1) + [E_e(t) + E_d(t)] \times \eta_c \tag{13}$$

Where, $E_b(t)$ is the stored energy of the BAT at any time t(kWh), $E_e(t)$ is excess energy supplied, $E_d(t)$ is the excess energy from the wind and solar PV system, η_c is the charging efficiency of the storage system.

In the discharging mode, the area's demand exceeded the available energy by renewable energy resources. Hence for the given hour t, the battery bank capacity is calculated using Eqs. (14)-(15) (Singh & Basak, 2021).

$$E_b(t) = (1-\propto) \times E_b(t-1) - \left[\frac{E_{nl}(t)}{\eta_c \eta_d}\right] \tag{14}$$

$$E_{nl}(t) = E_d(t) - [E_e(t) + E_d(t) \times \eta_c] \tag{15}$$

Where, $\propto$ is the hourly self-discharge rate, η_d is the discharging efficiency of the battery. E_{nl} is the net load, E_d is deficit in demand.

5. Optimization problem

Integrating multiple renewable energy sources is a challenging task. Because renewable energy sources are sporadic, the long-term output is unattainable. Designing a renewable power output and production variability that matches demand needs comprehensive planning. The model requires an estimate of the energy share of the supply inputs to lower the generation cost. It is essential to analyze the benefits and drawbacks of various resources while designing a microgrid model to satisfy varying load requirements. An integrated model has been developed for several villages in Uttarakhand, India. The suggested microgrid architecture's primary goal is to reduce total energy consumption and optimize component sizing. The model was designed in the MATLAB programming environment and implemented in a hybrid system to address design issues in remote areas.

Economic, technical, and environmental evaluations are three essential objectives that significantly impact microgrid development. These three goals have a significant impact on one another. Increasing the technological features, for example, considerably increases the cost of the microgrid while lowering the cost reduces the system's reliability. The current initiative aims to save energy costs by considering the possibility of a power shortage and implementing the most cost-effective electrification.

5.1. Objective function

The microgrid is designed to provide the lowest COE in the selected area. Thus, the chosen objective for the suggested study is to minimize the cost of electrification and provide reliable electrification of the area.

$$\mathrm{Obj\ F} = \min(\mathrm{TNPC},\ \mathrm{COE})$$

5.1. Cost of energy

The cost of energy (COE) ($/kWh) is measured by the average TNPC and can be expressed by Eq. (16) (Aziz et al., 2020).

$$\mathrm{COE} = \frac{(NPC)_T}{\sum_{N=1}^{8760} P_N} \times \varnothing_{cpv} \tag{16}$$

Where, NPC_T is the net present cost, P_N is the consumption of electrical power hourly, $\varnothing_{cpv}$ is the capacity recovery factor.

The total net present cost NPC_T is the sum of total investment cost, O&M cost, replacement cost, and fuel cost. Hence the total net present cost calculated by Eq. (17)

$$NPC_T = In_c + O\&M_c + R_c + F_c \tag{17}$$

The investment cost (In_c) is the sum of the initial cost of system components and mathematically can be expressed by the Eq. (18) (Hakimi et al., 2021).

$$In_c = N_S C_S + N_w C_w + N_H C_H + N_{bio} C_{bio} + N_{bg} C_{bg} + N_{con} C_{con} \tag{18}$$

Where, $N_S, N_W N_H N_{bio} N_{bg} N_{con}$ represents the optimum number of PV modules, wind turbines, hydro-turbine, biomass, and biogas generators and converters.$C_S,\ C_W,\ C_H,\ C_{bio}\ , C_{bg}, C_{con}$ refers to the cost of different components.

5.2. Deficiency of power supply probability (DPSP)

The reliability hybrid system is measured in terms of DPSP. The DPSP is a statical index representing the probability of power system failure. Due to the intermittency of resources may be low generation at the output or some technical loss. The range of values of DPSP is 0 to 1. The value zero implies the demand is fulfilled in all conditions and other values suggest the need is not fully satisfied. The DPSP is computed by the relation represented in Eq. (19) (Suman et al., 2021)

$$\mathrm{DPSP} = \frac{\sum_{n=1}^{n=t} (P_{gen}(n) - P_L(n)}{H_n} \tag{19}$$

P_{gen}, P_L and H_n are the power generated, power demand, and the number of hours, respectively.

5.3. Constraints

The multi-objective function of the given problem is subjected to satisfy the following constraints:

5.4. Power balance constraints

The power produced by renewable resources (PV, MHP, WT, Biogas, and storage) must not exceed the power system's operational restrictions to satisfy the system load requirement. At the specified load points, the electricity produced by the resources indicated must be sufficient to supply the area's load demand. At a particular instant, the demand of the area p_d must be satisfied by the power generated by different renewable resources and batteries specified by its maximum generating limits. The constraints represented by the Eq. (20) limits the power system.

$$\sum_{d}^{n_d} P_d(d,\partial) = \sum_{i=1}^{n_{Hps}} P_1(i,\partial) + \sum_{j=1}^{n_{PV}} P_2(j,\partial) - \sum_{k=1}^{n_{bat}} P_3(k,\partial) + \sum_{l=1}^{n_{bat}} P_4(l,\partial) + \sum_{p=1}^{n_{WT}} P_5(p,\partial) + \sum_{j=1}^{n_{BCG}} P_6(j,\partial) + \sum_{j=1}^{n_{BMS}} P_7(j,\partial) \quad (20)$$

5.5. Power generating output constraints

Each producing unit's power output is constrained to the individual original equipment manufacturers' maximum and minimum power restrictions. Below is a list of each resource's limitations.

$$\begin{cases} p_1^{min}(i,\partial) \le p_1^i(i,\partial) \le p_1^{max}(i,\partial) \\ p_2^{min}(j,\partial) \le p_2^j(j,\partial) \le p_2^{max}(j,\partial) \\ p_3^{min}(k,\partial) \le p_3^k(k,\partial) \le p_3^{max}(k,\partial) \\ p_4^{min}(l,\partial) \le p_4^l(l,\partial) \le p_4^{max}(l,\partial) \\ p_5^{min}(p,\partial) \le p_5^i(p,\partial) \le p_5^{max}(p,\partial) \\ p_6^{min}(m,\partial) \le p_6^m(m,\partial) \le p_6^{max}(m,\partial) \\ p_7^{min}(h,\partial) \le p_7^h(h,\partial) \le p_7^{max}(h,\partial) \end{cases} \quad (21)$$

$P_d(d,\partial), P_1(i,\partial), P_2(j,\partial), P_3(k,\partial), P_4(l,\partial), P_5(p,\partial), P_6(m,\partial), and P_7(h,\partial)$ is power demand, power output of HPS, PV, battery and wind turbine, BCG and BMS respectively.

5.6. Battery constraint

Battery bank cannot be overcharged or over discharged at any time t. So, order to operate within safe parameters, the battery bank has been subject to the following limitations:

$$E_{bat_{min}} \le E_{bat}(t) \le E_{bat_{max}} \quad (22)$$

Where $E_{bat_{min}}$ and $E_{bat_{max}}$ are the minimum and maximum capacity of the battery. The minimum and maximum storage capacity can be determined by the following equations (Han et al., 2019).

$$E_{bat_{min}} = \frac{n_{bat} v_{bat} c_{bat}}{1000} . SOC_{min} \quad (23)$$

$$E_{bat_{max}} = \frac{n_{bat} v_{bat} c_{bat}}{1000} . SOC_{max} \quad (24)$$

Where n_{bat}, v_{bat} *and* c_{bat} is the number of batteries, voltage of battery and capacity of the battery respectively. SOC_{min} and SOC_{max} is the maximum and minimum state of charge of the battery.

The size of the SPV, wind, and battery bank storage systems has been modified in the system under consideration to satisfy the electrical energy demand of the research region, while the MHP, biomass gasifier, and biogas-based generators' seasonal power capacity has been fixed. Upper and lower limits for wind, solar, and battery bank storage systems. Suggested microgrid for giving the rural community the most efficient, least-expensive electrification. Multiple technological and physical constraints are therefore taken into account during process optimization. The following limitations are therefore taken into account for optimization:

$$N_{S,min} \le N_S \le N_{S,max}$$

$$N_{W,min} \le N_W \le N_{W,max}$$

$$N_{bio,min} \le N_{bio} \le N_{bio,max}$$

$$N_{H,min} \le N_H \le N_{H,max}$$

$$N_{bg,min} \le N_S \le N_{bg,max}$$

Where, $N_{S,min}$, $N_{W,min}$, $N_{bio,min}$, $N_{H,min}$, $N_{bg,min}$ are the minimum number of solar modules, wind turbine, biomass generators, hydro turbines, and biogas generators. Similarly, $N_{S,max}$, $N_{W,max}$, $N_{bio,max}$, $N_{H,max}$ and $N_{bg,max}$ denotes the maximum number of components. Further renewable fraction is also a constraint in the optimization.

5.7. Differential evolution (DE)

The stochastic evolutionary search method known as DE is reliable. It has been effectively tested on numerous well-known benchmark functions as well as nonlinear, non-convex, and non-differentiable optimization issues in the real world. R. Storm and K. Price advocated for differential evaluation in the 1996 and 1997 publications (Das & Suganthan, 2011). DE, like pattern search (PS) and GA, is a vector-based optimization technique due to mutation and crossover. DE is a stochastic search method that does not rely on derivative information and tends to self-organization. The method does away with encoding and decoding and is population-based and derivative-free. DE is an easy-to-use, straightforward algorithm (Moradi et al., 2015).

The results show that DE and its derivatives significantly outperform other evolutionary techniques. DE was used to address a range of problems. A DE technique is used to create trial vectors from a collection of starting populations. Through the addition of weighted random vector differentials, DE modifies vectors at each step (Gao et al., 2021). The trial vector replaces the target vector when its fitness in the following generation is higher than that of the target vectors. DE is applied to numerous optimization issues. The DE algorithm's functioning is divided into four parts (Alam et al., 2016). The algorithm begins with the initialization stage, which generates an initial population. Second, the algorithm introduces mutation into the starting population. Third, a crossover is conducted, followed by a selection to establish a population. The system iteratively repeats the mutation process until the termination condition is reached (Wang et al., 2019).

5.8. Initialization

The Differential Evolution optimization algorithm seeks to increase a population N_P of individuals, or viable solutions, in d-dimensional parameter vectors i.e., $Y_{i,g} = \{Y_{i,g}^1, Y_{i,g}^2, Y_{i,g}^3 ..... Y_{i,g}^d\}$ where, i=1,2,3….N_P, in the path of the global optimal. The initial population should include as much of the search space as is feasible by uniformly randomly selecting individuals from it. The boundaries of the search space's maximum and minimum parameters are illustrated as:

$$Y_{min} = \{Y_{min}^1, Y_{min}^2, Y_{min}^3 ..... Y_{min}^d\} \quad (25)$$

$$Y_{max} = \{Y_{max}^1, Y_{max}^2, Y_{max}^3 ..... Y_{max}^d\}.$$

For instance, the eq. (26) at the generation where g=0 creates the initial value of the parameter of the individual (Mandal & Mandal,

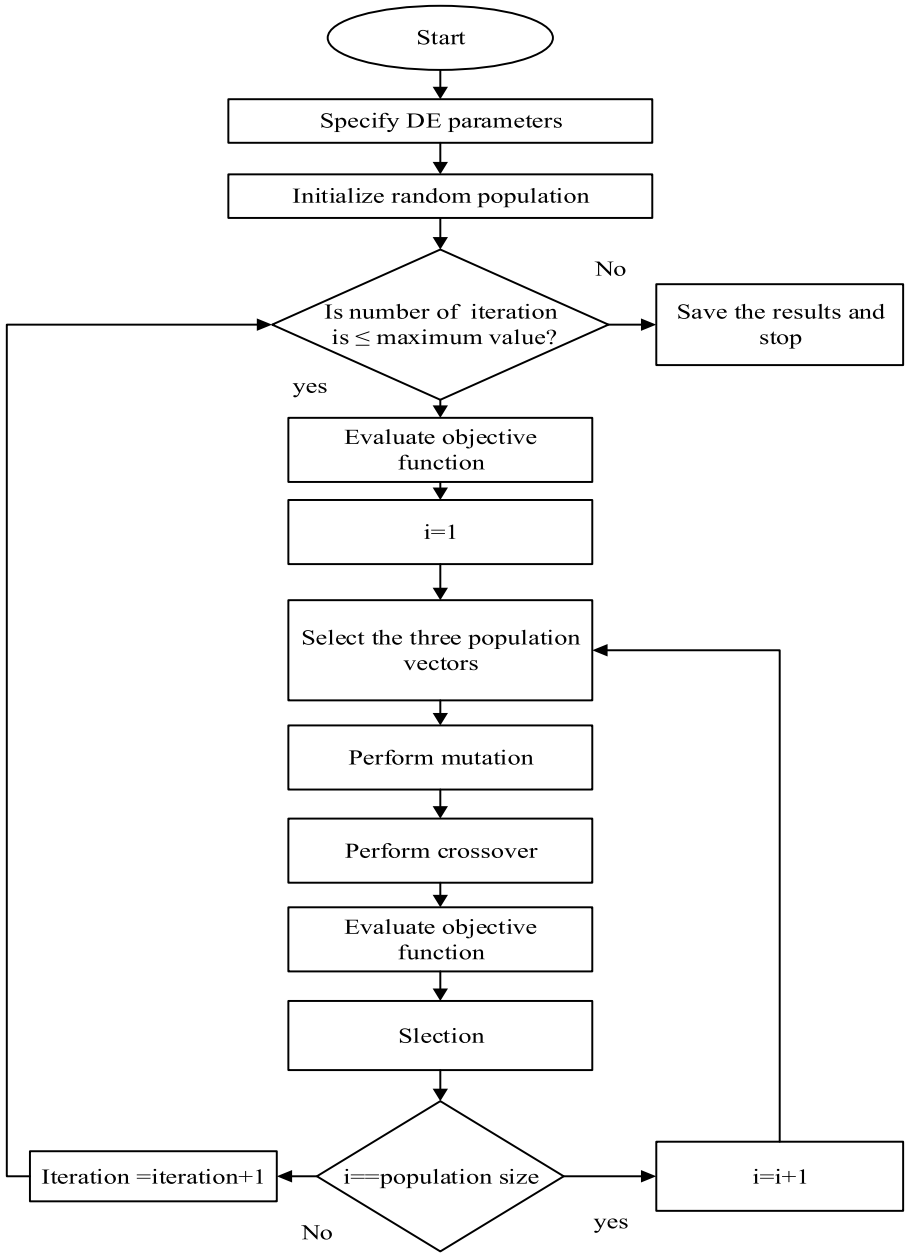


Fig. 10. Flowcharts for DE algorithm.

2020).

$$Y_{i,0}^{m} = Y_{i,0}^{m} + rand(0,1) \times \left(Y_{max}^{m} - Y_{min}^{m}\right), \mathrm{m} = 1, 2\ldots\ldots, \mathrm{d} \tag{26}$$

Whereas rand (0,1) designates a randomly distributed variable with uniform distribution, denotes a randomly distributed variable with a uniform distribution that fluctuates between (0,1).

5.9. Mutation

The mutation approach includes beginning after the initialization of parameters. Numerous mutation techniques have been documented in the literature. The most critical integrant in the mutation process is the vector difference. The vector difference is a crucial component in the mutation process. The mutant vectors (V_i) are produced by mutant operators(Y_k) by perturbing a random vector with the difference between two additional chosen vectors (Y_i and Y_n) provided by the Eq. (27).

$$V_{i=}^{g}\ Y_{k}^{g} + f_s\left(Y_{i}^{g} - Y_{m}^{g}\right) \tag{27}$$

Where f_s is a scaling factor that falls between [0,2] and the three vectorsY_k^g, Y_i^g, Y_m^g are chosen at random.

5.10. Crossover

The crossover operation led to an increase in mutant vector variety. Following mutation, the crossover was employed to produce the trail vectors for each pair of the target vector Y_i^g and matching mutant vector V_g^i.

$$U_{i,g}^{j} = \left(U_{i,g}^{1}, U_{i,g}^{2}, U_{i,g}^{3}\ldots\ldots\ldots U_{i,g}^{d}\right) \tag{28}$$

The uniform crossover is used in the basic version of DE, specified by the Eq. (29)

Table 2
Parameters of components used in modeling.

PV (Adefarati & Bansal, 2019)		Biomass (Azerefegn et al., 2020)	
Capital cost	630$/kW	Capacity	40kW
Capacity	300 W	Capital cost	$ 23,700
Replacement cost	630$/kW	Replacement cost	$ 18000
Lifetime	25 years	O& M cost	0.05/ hour
O&M cost	10$/kW	Lifetime	15,000 h
Wind (Azerefegn et al., 2020)		Biogas (Suman et al., 2021)	
Rated power	10kW	Capital cost	$ 385 /kW
Capital cost	$ 35,000	Capacity	20kW
Lifetime	25 years	Replacement cost	$ 335/ kW
Replacement cost	$ 30,000	O&M cost	0.2 $/kW
O&M cost	1000/year	Lifetime	15000 h
Speed	V_c=3m/s V_o=20m/s	Converter (Adefarati & Bansal, 2019)	
Hub height	20m	Capital cost	300 $/kW
Battery (Alshammari & Asumadu, 2020)		Lifetime	15 years
Capacity	4kWh,84Ah	O&M cost	3$/kW
Capital cost	300$/battery	Replacement cost	300 $/kW
Replacement cost	300$/battery	Micro-hydro (Chauhan & Saini, 2016c)	
O&M cost	104$/battery/ year	Capacity	30 kW
Lifetime	5 years	Capital cost	1333 $/kW
SOC	Max=95%, min=40%	Replacement cost	1333 $/kW
DOD	30%	O&M cost	800$/yr.
Charging efficiency	85%	Lifetime	25 years

$$U_{i,g}^{j} = \begin{cases} V_{i,g}^{j} \ if \ rand_j[0,1] \le C_R \ or \ J = j_{rand} \\ Y_{i,g}^{j} \ otherwise \ j = 1,2,3.....d \end{cases} \quad (29)$$

Where C_Ra user-defined variable with a range of [0,1], is the crossover rate. The crossover rate constant regulates how much information is translated from mutant vectors into parameter values. The binomial crossover rate copies thej^{th} parameter of the mutant vector to the equivalent element in the trailing vector.

5.11. Selection

We typically initialize a trail vector parameter inside the expected range if its value falls outside of the upper and lower bounds. All trail vectors' goal functions are investigated. The selection operation follows the crossover procedure. To evaluate the objective function, the appropriate target vector f ($Y_{i,g}$) is compared to each trail vector f ($U_{i,g}$) . If the objective function value is less than or equal to the value of the correlated target vector, the trailing vector will replace the target vector in the population of the subsequent generation; otherwise, the vector will stay in the population. The following are the selection vectors:

$$Y_{i,g+1} = \begin{cases} U_{i,g} \ if \ f(U_{i,g}) \le f(Y_{i,g}) \\ Y_{i,g} \ otherwise \end{cases} \quad (30)$$

Unless it is replicated from generation to generation, the process ends once a termination requirement is met. The DE algorithm's flowchart is depicted in Fig. 10 (Rana & Lalwani, 2017).

6. Case study

The proposed microgrid model for rural community electrification is the subject of a case study. As discussed in the case study, a remote rural region of Uttarakhand (India) is selected. The model is built by integrating the area's available energy resources. The suggested model includes hydro, solar PV, biomass, batteries, and a biogas-producing unit. The technical and economical specification of the components used in the modelling shown in Table 2.

The suggested work compared various optimization techniques in the MATLAB environment and used them to solve the problem that was created for the case study. Optimizing optimization techniques (GA, PSO, and DE) optimize the overall energy cost. The various criteria for the optimization algorithm are displayed in Table 3. The starting population and stopping conditions for GA, PSO, and DE remained the same throughout the performance comparison. GA typically uses a low mutation rate of less than 0.2 and a crossover operation between 0.6 and 1.0.

Table 3
Various optimization techniques' parameters.

Optimization algorithms	Iteration	Control parameters
GA (Sanajaoba & Fernandez, 2016)	100	S_f=0.80, M_r: 0.20, selection function: Roulette
PSO (Askarzadeh, 2017)	100	w_{max}=0.90, w_{min}=0.4, C_1=2.8, C_2=2.0
DE (Kamal et al., 2022)	100	W^f=1.8, C_p=0.8

7. Results and discussion

As previously stated, this research aims to provide an optimal, integrated framework of a renewable energy system for a remote rural area where grid infrastructure is inadequate. The literature uses a variety of evolutionary approaches to meet energy demand while minimizing energy costs. The recommended configuration of the microgrid model takes into account the large renewable energy resources in the area. Optimization approaches (GA, PSO, and DE) were utilized to evaluate the model built by integrating locally accessible energy resources. To offer the optimum model for electrifying the region, the findings of the minimal optimized value of the COE are investigated. While modeling the energy system, the intermittency of renewable energy resources is considered. Based on the above classification, different case has been discussed. The proposed model was implemented in the MATLAB simulation environment and provided energy via an independent microgrid system comprised of solar PV, BG, WES, MHP, BM, and BAT. The following case is considered for analysis and feasibility study.

The various combinations of renewable energy resources are optimized for the least TNPC and COE using PSO, DE, and GA optimization techniques. The three algorithms are tuned for the least operation cost. Fig. 11 shows the convergence curve for Case:4. Fig. 12 displays the variation of the TNPC and COE by three algorithms. The close observation of the figure suggests that for all the case, the results provided by the DE algorithm is more optimal. Hence for further analysis, DE algorithm-based results are used.

The suggested microgrid configuration is analyzed using the DE algorithm for the feasible model suggestion for the least cost of the electrification of the area.

Case: 1 consists of MHP/BG/BM and BAT to electrify the community. Storage units are used in the arrangement to increase the reliability of the supply. Simulation results suggest that the configuration includes a 40kW MHP, a 30kW BG, a 30kW BM gasifier system, a 54 kWh BAT unit, and a 6kW converter to meet the energy need of the area. The TNPC of this configuration is $ 5,28,986, and 0.21 $/kWh is the value of COE. The arrangement contains mainly renewable-based resources and storage units. The production of power from renewable resources has many external constraints. Hence the energy produced is not a reliable source of electrification. The suggested system cannot meet the area's demand, and an energy shortage of 25% is reported at the user end. Case: 2

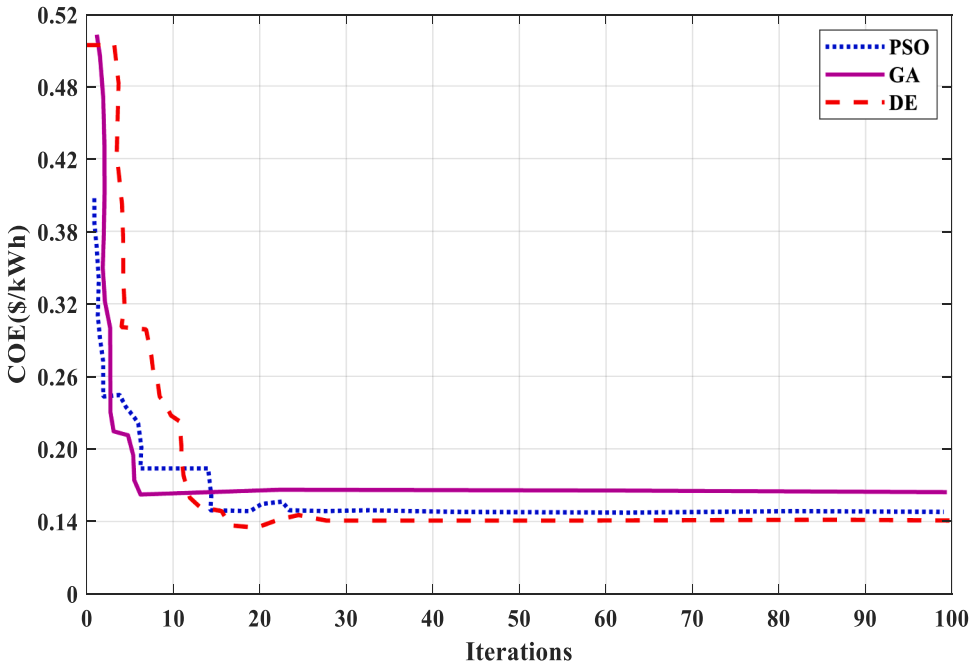


Fig. 11. Convergence curve for the case:4.

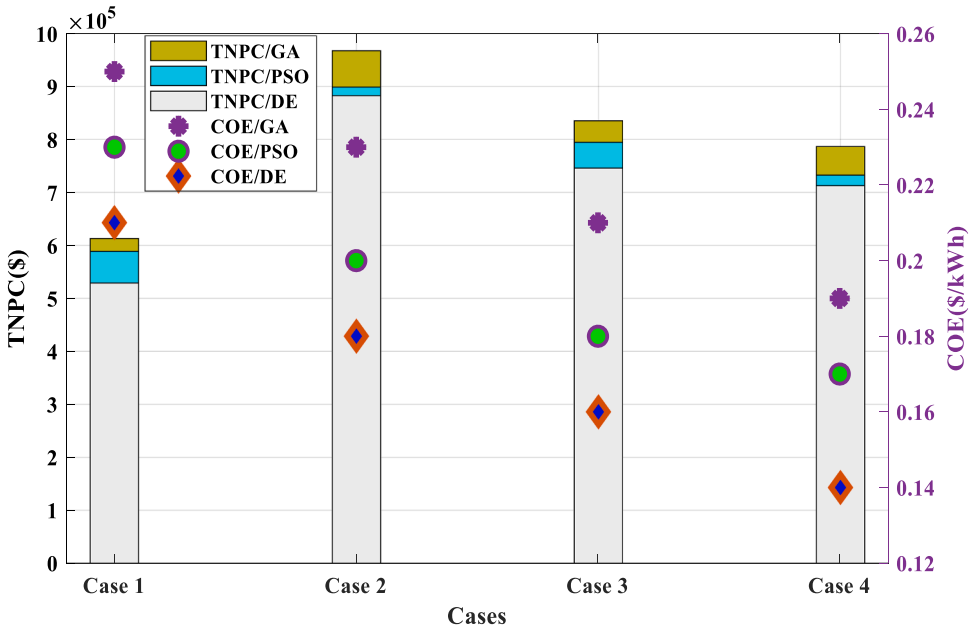


Fig. 12. Variation of TNPC and COE using the optimization algorithms.

consists of MHP/BG/BM/WT and BAT.

The model includes a 40 kW MHP, 30 kW BG, 30 kW BM gasifier, 92 kW WT, 1431 kWh BAT, and 52 kW converter. WT was added to the system, resulting in a 40% increase in the system's cost. The proposed system has a COE of $0.18 per kWh and a TNPC of $8,82,532, resulting in a COE of $0.18/kWh. The scheme is capable of 100% load satisfaction. However, the system's TNPC and COE are increasing, making it unviable. MHP/BG/BM/SPV and BAT are used in the Case: 3 configurations. The solar PV system is added to improve the reliability of the power supply. A 40 kW MHP, a 30 kW BG, a 30 kW BM, a 92 kW SPV, a 638 kWh BAT unit, and a 52-kW converter make up the best combo design. This combination's TNPC and COE are calculated to be $ 7,45,753 and $0.16 / kWh, respectively. The energy generated by a solar photovoltaic system is highly variable. The increased total net present cost (16%) for a capacity deficit of 0% at the user end correspondingly. The system has the least cost compared to previous cases but has no reliable electrification method due to fewer sources in the model.

Case: 4 consists of MHP/BG/BM/WT/SPV and BAT. The system combines all the significant renewable energy resources of the area. The combination increased the reliability of the power supply due to a reduction in the dependence on one or more renewable resources. When there is an excess of energy production compared to demand and supply or even when one or more sources are not operating, the battery system is available to store the power. The optimum configurations for the system are a 40 kW MHP system, a 30 kW BG, a 30 kW BM, a 40 kW WT, a 60 kW SPV, a 329 kW BAT unit, and a 52-kW converter. The TNPC and

Table 4
Simulation results of various cases by DE algorithm.

Cases	SPV (kW)	WT (kW)	MHP (kW)	BG (kW)	BM (kW)	BAT (kWh)	CON (kW)	TNPC ($)	LCOE ($/kWh)	Shortage (%)
Case:1	-	-	40	30	30	54	6	528986	0.21	25%
Case:2	-	87	40	30	30	1431	52	882532	0.18	0%
Case:3	93	-	40	30	30	638	52	745753	0.16	0%
Case:4	60	43	40	30	30	329	52	712532	0.14	0%

Table 5
Cost breakdown for case 4.

Components	Capital cost ($)	Replacement cost ($)	O&M cost ($)	Fuel cost ($/yr.)
MHP	56778.76	0.00	24168.75	0.00
BG	29890.00	41673	22456	36898
BM	39325.65	84542	18664	44243
WT	54461.00	0	21093	0
SPV	75981.24	0	18945	0
BAT	49642	151657	12454	0
CON	5387	3684	1741	0

COE for this combination were calculated to be $7,12,532 and $0.14 per kWh, respectively. The suggested system can handle the entire demand in the area and has no capacity shortages at the user end.

Various microgrid models are considered for optimal component sizing and the lowest cost of operation. Table 4 shows a number of microgrids with different component configurations. The DE algorithm is used to optimize the configuration outcomes. The above renewable resource combination of microgrids attempts to meet the area's electricity needs. Because renewable-based generation is highly intermittent, all microgrids are not viable electrification solutions for the area. In other circumstances, energy shortages have been recorded as a result of uneven production. The demand deficit is depicted in Table 4. The following combinations are insufficient to meet the entire area's needs. The rural grid's development considers the intermittent nature of renewable energy and supplies reliability. The various component combinations are tested to see which one has the lowest cost of electrification. The best microgrid may be created in the scenario described above by integrating all local resources. This reduces the dependency of electricity supply on single resources. Hence, the combination is formed by the 40 kW MHP system with 30kW BG, 30kW BM gasifier system, 60 kW PV system, and 43kW of WES. The TNPC of the combination is $ 712532 and 0.14 $/kWh, respectively. For the modeling of demand of power supply probability 100%.

For the DSPS=0 and feasible microgrid formed with micro-hydro, solar, biomass, biogas and battery system. Component-wise cost distribution of the microgrid is shown in Table 5.

Storage units are an integral part of the designed system. The storage unit smoother the power supply and increases the supply's stability. The battery units supply the power when one or more energy resources are not working, or due to inadequate environmental conditions, they are not operational. Fig. 13 shows the BAT unit's annual variation of the state of charge (SOC). For the analysis of the current microgrid, two seasons are considered. Fig. 14 shows the variation of SOC of the battery for two seasons.

The best-case scenario can meet the area's energy needs while producing electricity entirely from renewable resources. Fig. 15 shows the contribution of energy produced by the various components. Fig. 15 suggests that a significant portion of the energy supplied by the MHP (42%) is supplied.

8. Sensitivity analysis

Several factors have an impact on the microgrid components that are employed in energy production. The crucial parameter for change in the suggested system considers wind speed and ambient solar radiation. Given that it illustrates how changing input parameters affect microgrid planning, the suggested model merits further study. These variables directly impact the system's overall output. The following justifies the selection of these parameters for analysis. Market factors impact oil prices, resulting in price increases or decreases. Second, the sun's radiation and wind speed are affected by meteorological factors (Fig. 16).

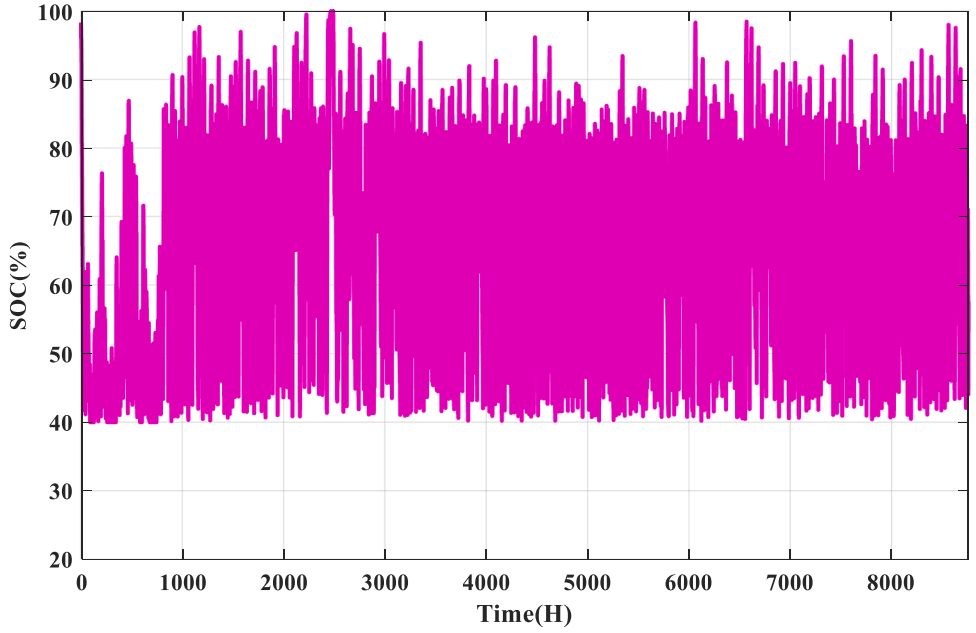


Fig. 13. SOC of the battery.

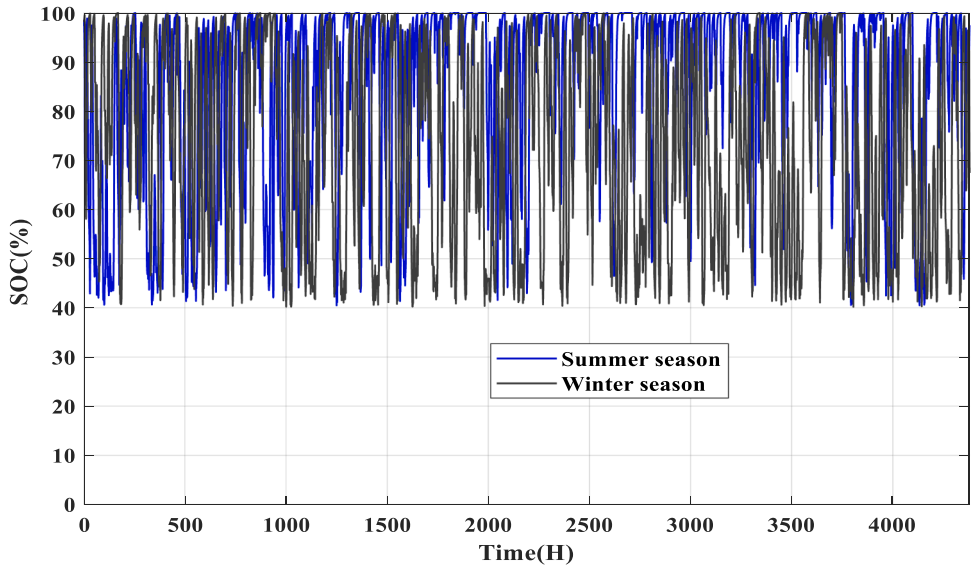


Fig. 14. Seasonal variation of SOC of the battery.

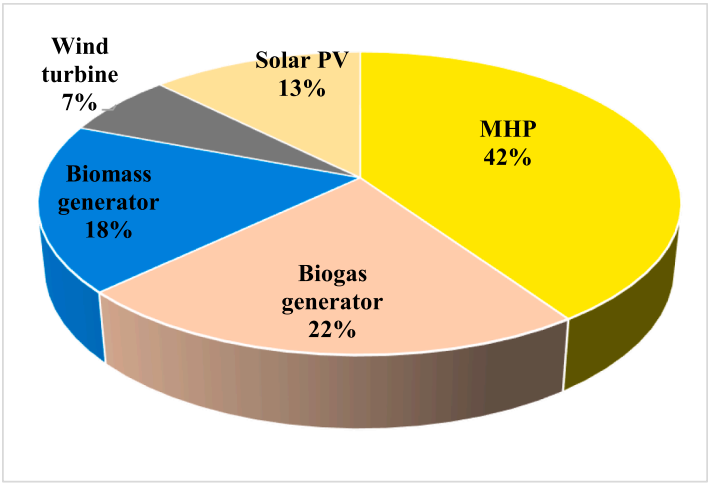


Fig. 15. Contribution of various resources.

The abscissa of the plot depicts parameter variation, whereas the y-axis depicts COE variation. COE has a higher sensitivity, as demonstrated by its ability to detect 70% fluctuations in wind and solar generation units. As a result, the COE grows as the quantity of solar and wind generation units decreases. When the quantity of SPV and WT is increased, the COE rate drops dramatically.

As the quantity of renewable resource units in the system grows, the fluctuation becomes more stable and cost-effective. Renewable energy systems have a longer life duration than other microgrid components, which is why they are more cost-effective in the long run. The number of biomass producers has reduced COE to a 35% variation; as the variation grows, so does the cost of operation, making energy more expensive. The sensitivity of load change reduced the COE to 25% as more load was added to the system. The cost of energy climbs after that. Fig. 16 depicts the sensitivity profile of several components.

Fig. 17 depicts the sensitivity analysis of the ideal system with a rate of interest. The interest rate varied from 6% to 12%, and the consequence of this variation on the TNPC and COE is investigated. The overall net present cost of the system increases when the interest rate rises, as shown in Fig. 17, with the value of COE increasing from 0.14 $/kW h for a 6% interest rate to 0.22 $/kWh at a 12% rate of interest. The COE lowers by 18.63% when the interest rate is cut in half, as shown in Fig. 17.

Fig. 17 shows that as the interest rate of the system increases, the COE and TNPC also increase. As a result, an optimal interest rate is required for the best operation of the ideal microgrid system.

The initial cost of PV is a significant consideration when designing a microgrid. Geographical location affects the initial cost. Fig. 18 shows the impact of increasing the PV initial cost from $330/$m^2$ to $730/$m^2$ on the TNPC and COE. For a $330/$m^2$ PV starting cost, the COE increases to 0.20 $/kWh for a $730 /$m^2$ PV starting cost. Similarly, the decrease in COE is reflected in the lower beginning cost. The COE is decreased by 12% when the starting cost is reduced by 40%.

Biogas is the primary source of energy for rural people. Fig. 19 depicts the sensitivity analysis results for the biogas system's initial cost. For analyzing the impact of variation of initial cost on TNPC and COE, the initial cost varied from $3500 to 8000/kW. The COE for a biogas system increases from an initial value of 0.134 $/kWh to 0.22$/kWh for the variation in the range from $3500/kW to 8000 $/kW. When the initial cost of the biogas system is decreased by 45%, the COE is reduced by 32%.

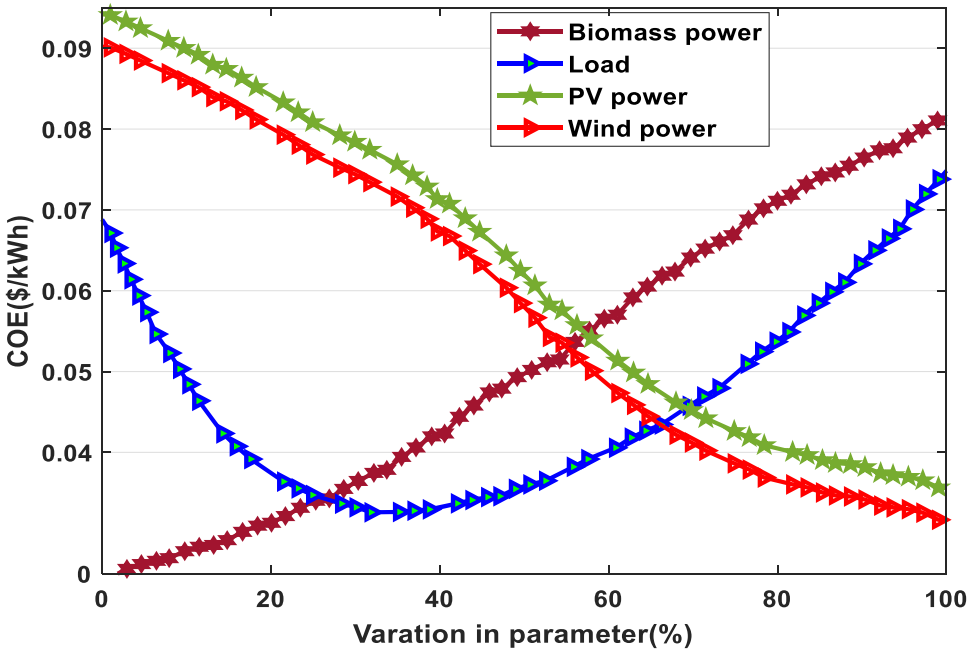


Fig. 16. Sensitivity of the components.

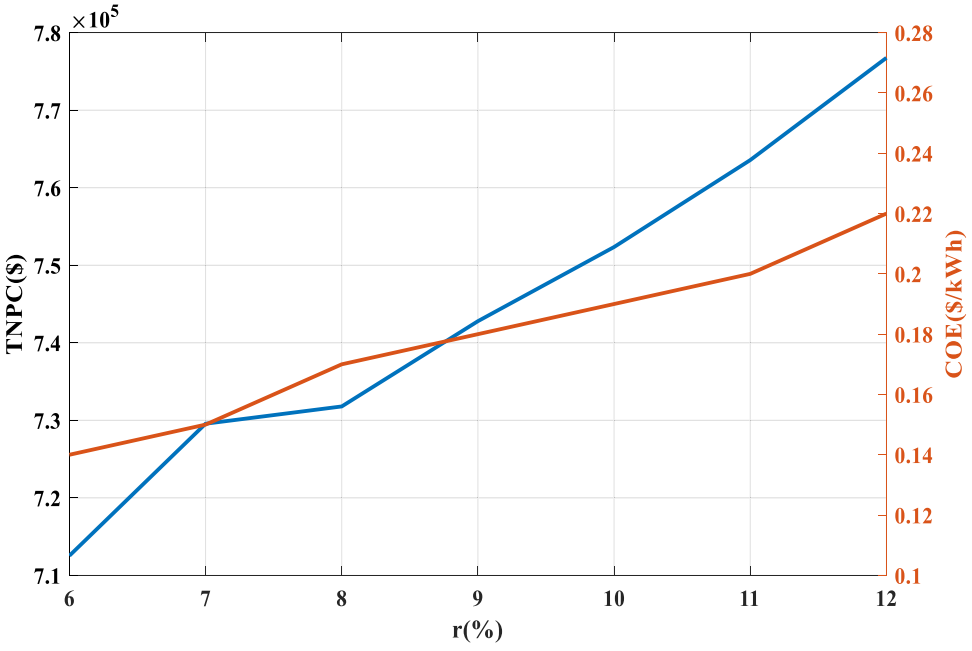


Fig. 17. Variation of COE and TNPC with r.

9. Comparison with existing literature

The suggested microgrid model optimized the COE and NPC values with the least COE and NPC. The results optimized component sizing and investigated the impact of the parameter change on COE. The collected results are compared to existing results in the literature for results validation. Table 6 suggests comparing the findings to recent research in a similar field. Comparing the detailed study in terms of load profile, optimization approach, and similar parameters is not possible, and a fair comparison can be possible in terms of an economical approach. Hence for fair compassion with a similar recent study, the comparison in terms of COE and TNPC is carried out.

For the efficient electrification of the rural community of Egypt, a microgrid was presented by Eteiba et al. (2018) for an average load of 165kW. The optimal sizing of components and minimizing the COE and TNPC were performed using FPA, ABC, and FA. The optimal solution suggests that the microgrid has an LCOE of 0.144/kWh and a TNPC of 38,58,114. The present study system has few renewable resources, i.e., PV/BG, and a battery is provided for energy storage, which has less reliability and a high TNPC value. Similarly (Bhatt et al., 2016) studied the 5 villages of Uttarakhand(India) for an average load of 675.78kW/day. The model consists of MHP/PV/BG/BM/WT. The COE of the system is 0.197$/kWh, and the total NPC is $ 5,33,654. The present study system has no storage unit and a higher value of COE. In another study (Misra & Sharma, 2020) of Kutch (India), the optimal system suggests a higher TNPC value ($ 8,30,440) compared to the

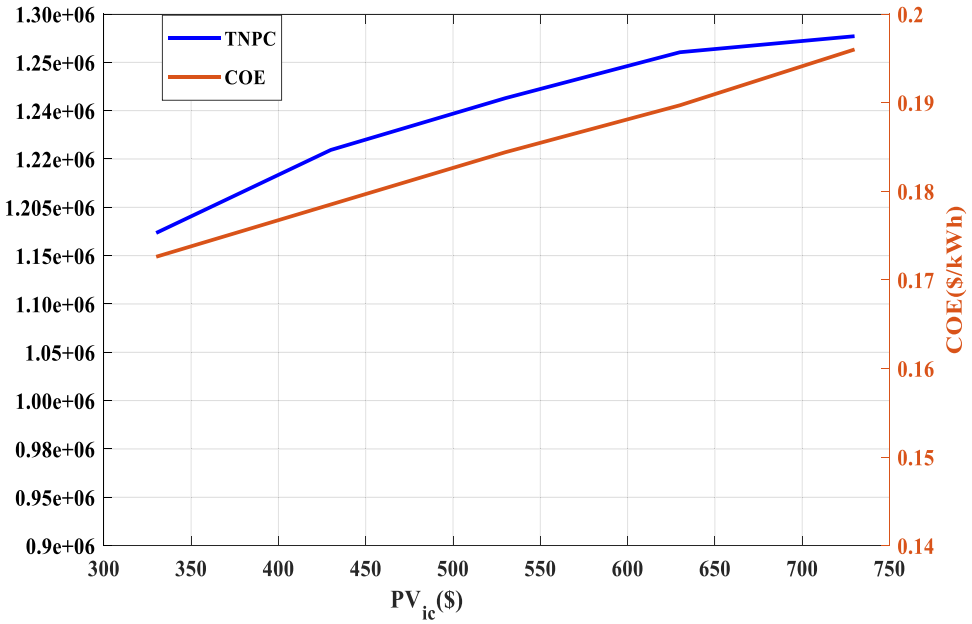


Fig. 18. Sensitivity with PV cost.

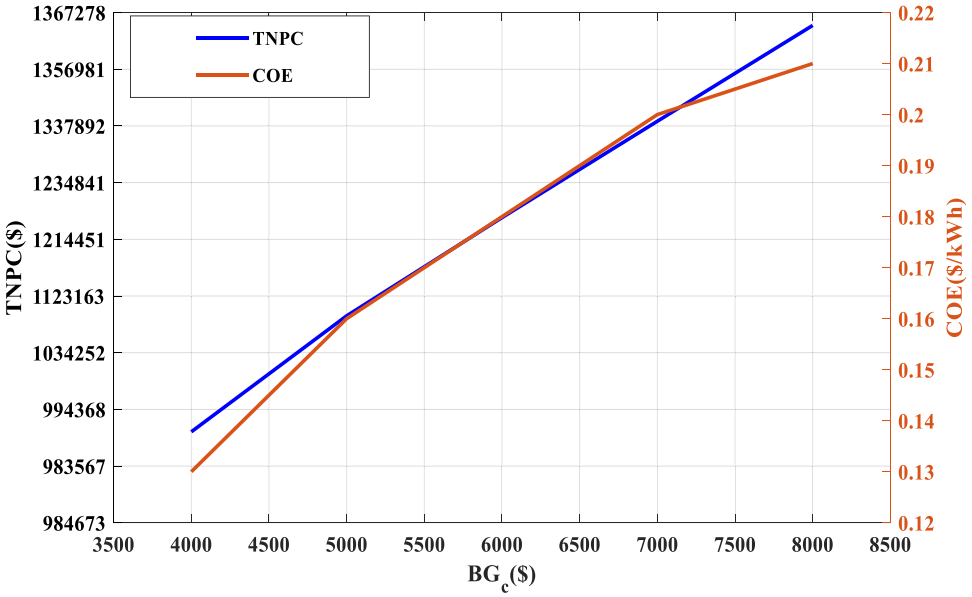


Fig. 19. Sensitivity of biogas price.

present study. The existence of DG units discharges more toxins into the environment. Similarly, the studies in Table 6 show that they either have higher COE and TNPC values comparable to the present study. The comparative analysis suggests the present investigation is more feasible compared to the previous study.

10. Conclusion

This study develops a hybrid energy model using domestically obtained energy sources such as SPV, HPS, WES, BGS, BMS, and battery. The current initiative attempts to provide the area with reliable electricity. The integrated concept is being tested for a specific Indian rural community in Uttarakhand (India). To lower TNPC, COE, size optimization, sensitivity analysis, and techno-economic viability optimization techniques are applied. The metaheuristic algorithm DE is used to size the proposed microgrid system. The microgrid is established at zero percent DPSP and a minimum COE value to meet the villagers' demands. According to the findings, the DE can acquire the global optimal in a small amount of time. The best system configuration consists of SPV, WES, MHP, BM, BG, and BAT. The net present value of the combination is $712532, and COE is 0.14 $/kWh. For the modeling demand for power supply, the likelihood is 100%. The optimized results are compared to the GA and PSO outcomes.

It may be concluded that the strategy used can assist in gaining insight into the technical aspects of developing microgrids with smaller power supply shortfalls for remote places. Furthermore, the economic aspects of the various merger of renewable energy resources can be defined, enabling the development of a sustainable energy model in economically disadvantaged cultures. Future research could include a fault-bearing capacity analysis of optimal combinations and other voltage and frequency variation options during grid integration.

Table 6
Comparison of results with the previous study.

Reference	System	location	COE ($/kWh)	Status	TNPC ($)
(Ribó-Pérez et al., 2021)	SPV/BM	Zambia	0.48	Off-grid	5,64,697.00
(Eteiba et al., 2018)	SPV/ BM/BAT	Egypt	0.14	Off-grid	38,58,114.00
(Rajbongshi et al., 2017)	BM/ SPV/DG/ BAT	India	0.11	Off-grid	1,61,169.00
(Hossain et al., 2017b)	SPV/ WES/ DG/BAT	Malaysia	0.28	Off-grid	1,71,50,000.00
(V. & Verma, 2021)	SPV/ BM/ BAT/DG	India	0.15	Off-grid	70,5000.00
(Bhatt et al., 2016)	SPV/ BAT/ MHP/ BM/BG	India	0.19	Off-grid	5,14,034.00
(Heydari & Askarzadeh, 2016)	SPV/BG	Iran	0.19	Off-grid	16,72,296.00
(Sigarchian et al., 2015)	SPV/ WES/ BG/ BAT	Kenya	0.25	Off-grid	1,17,700.00
(Misra & Sharma, 2020)	SPV/ BM/ DG / BAT	India	0.13	Off-grid	8,30,440.00
This study	SPV/BG/ MHP/ BM/BAT	India	0.14	Off-grid	71,2,532.00

Declaration of Competing Interest

The authors declare that they have no known conflict of interest.

Data availability

There are no scientific datasets used in this investigation. Information, facts, and figures are all cited and referenced in the current work.

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